

Microbial Discovery for PFAS Bioremediation

A 12-Month Proposal for Identifying and Characterizing
Native Bacterial Strains Capable of C–F Bond Cleavage

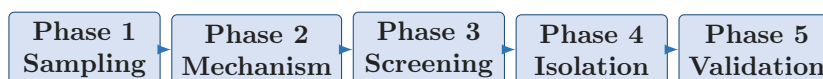
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Abstract. Per- and polyfluoroalkyl substances (PFAS) are a family of synthetic chemicals notorious for their environmental persistence; the C–F bond is among the strongest single bonds in organic chemistry, and remediation today relies heavily on energy-intensive physical methods such as granular activated carbon adsorption and high-temperature incineration. This proposal outlines a 12-month research program to discover, isolate, and characterize native microbial strains capable of degrading PFAS *in situ*. We will sample contaminated water and biomass from 4–6 sites identified by **PFASolve**, screen enrichment cultures for fluoride release as a direct readout of C–F cleavage, isolate responsible organisms, and validate degradation under controlled laboratory conditions.

Project Overview

The five phases below progress from environmental sampling through validated proof of concept. Each phase is gated by a clearly defined go/no-go criterion so that resources can be redirected quickly if a phase fails to produce viable candidates.



Total duration: 12 months **Host institution:** Stanford University **Primary deliverable:** Validated PFAS-degrading isolate(s)

1. Phase 1 — Environmental Sampling

Phase 1 Environmental Sampling *Months 1–2*

Collect PFAS-contaminated waters and surrounding biomass from **4–6 locations** identified by PFASolve. Preserve samples cold (4°C, dark) for downstream cultivation and at –80°C to preserve DNA for later metagenomic analysis.

1.1 Site Selection Criteria

Sites will be prioritized in collaboration with PFASolve based on:

- Documented PFAS contamination above 70 ng/L (EPA health advisory level for PFOA/PFOS combined).
- Diversity of PFAS profiles — ideally a mix of long-chain (PFOA, PFOS) and short-chain (PFBA, PFBS) sites.
- Diversity of source types: AFFF-impacted military/airport sites, fluorochemical manufacturing outflows, landfill leachate, and at least one wastewater treatment plant effluent.
- Site accessibility and existing characterization data (reduces analytical overhead).
- At least one “low contamination” control site for baseline microbial community comparison.

1.2 Sample Types Collected at Each Site

- **Surface water:** 2 L per site, collected into pre-cleaned amber HDPE bottles.
- **Sediment:** 200–500 g from the top 10 cm, collected with a methanol-rinsed scoop.
- **Biofilm / attached biomass:** swabs and scrapings from submerged surfaces where present.
- **Rhizosphere soil:** from plants growing in or adjacent to contaminated water.

1.3 Field Logistics and Preservation

Samples will be transported on ice within 24 hours of collection. On receipt, each sample is split into three aliquots: (i) a working aliquot at 4°C in the dark for cultivation, (ii) a DNA-preservation aliquot at –80°C, and (iii) a chemistry aliquot reserved for quantitative PFAS analysis by LC-MS/MS to confirm contamination levels and provide a baseline for later degradation measurements.

Go/No-Go criterion (end of Month 2): At least 4 sites successfully sampled with confirmed PFAS contamination above detection limits and viable microbial communities (visible turbidity in initial enrichment within 2 weeks).

2. Phase 2 — Biological Mechanism

Phase 2 Biological Mechanism () *Months 2–4*

Two parallel enrichment experiments per sample: **(1)** Bacteria + PFAS only — identifies samples in which PFAS serves as a sole carbon and/or energy source. **(2)** Bacteria + PFAS + alternate carbon source (glucose, acetate) — identifies samples that co-metabolize PFAS as a side reaction while growing on a more accessible substrate. No-bacteria and heat-killed controls are run in parallel.

2.1 Rationale

The two-track design captures the two most likely metabolic strategies. PFAS as a sole substrate selects strongly but may yield no growth at all if no organism can use PFAS as a primary energy source. Co-metabolism is biologically more plausible for many recalcitrant compounds and is widely documented for related halogenated pollutants, but requires careful controls to distinguish actual PFAS degradation from incidental sorption.

2.2 Experimental Design

Each environmental sample is split into four conditions, run in triplicate:

Condition	Inoculum	PFAS	Co-substrate	Heat-killed?
1. PFAS only	live	yes	no	no
2. PFAS + glucose	live	yes	yes	no
3. No-bacteria control	none	yes	yes	n/a
4. Heat-killed control	killed	yes	yes	yes

2.3 PFAS Mixtures

Each condition is run with two PFAS spike profiles: a long-chain mix (PFOA, PFOS at 1–10 mg/L) and a short-chain mix (PFBA, PFBS at 1–10 mg/L). Concentrations are deliberately elevated above environmental levels to ensure detectable degradation signal during enrichment.

2.4 Readouts

- **Growth:** OD₆₀₀ measured every 24 h.
- **PFAS depletion:** LC-MS/MS at days 0, 14, 28.
- **Fluoride release:** ion-selective electrode at days 0, 14, 28 (a positive signal here is the key indicator that C–F bonds are actually being cleaved rather than the molecule simply sorbing to biomass).

Go/No-Go criterion (end of Month 4): At least one sample shows fluoride release above both no-bacteria and heat-killed controls by a factor of $3\times$ or more (or $p < 0.05$ across triplicates), indicating biotic C–F cleavage.

3. Phase 3 — High-Throughput Screening

Phase 3 High-Throughput Screening *Months 4–6*

Use a colorimetric fluoride-release assay to measure C–F cleavage activity across 96 cultures simultaneously. Confirm hits from Phase 2 with quantitative ion measurement.

3.1 Why Fluoride Release Is the Right Readout

Many candidate degradation mechanisms produce intermediate metabolites that look like degradation in LC-MS but actually represent biotransformation of one PFAS into another (transformation, not mineralization). Released fluoride is unambiguous: a fluorine atom in solution as F^- came from a C–F bond. Fluoride release is therefore the cleanest single readout for genuine degradation activity.

3.2 Assay Format

- **Plate format:** clear-bottom 96-well plates.
- **Colorimetric reagent:** a lanthanum–alizarin complexone system or equivalent fluoride-sensitive dye, read at 620 nm.
- **Throughput:** approximately 8 plates per week per technician, allowing diversification across PFAS substrates, dilutions, and time points.
- **Confirmation:** all colorimetric hits are re-tested with a fluoride ion-selective electrode for quantitative confirmation.

3.3 Screen Layout

Each plate includes the following columns: blank, no-bacteria control, heat-killed control, positive fluoride standard, plus 8 columns of biological samples in triplicate. Hits are defined as wells where fluoride concentration exceeds 3σ above the matched no-bacteria control.

Go/No-Go criterion (end of Month 6): At least 5 culture wells confirmed fluoride-positive by both colorimetric assay and ion-selective electrode, ready for single-colony isolation.

4. Phase 4 — Isolation and Genomics

Phase 4 Isolation and Genomics *Months 6–9*

Isolate single colonies from fluoride-positive wells, retest each isolate independently for fluoride release, and sequence the genomes of confirmed positive isolates.

4.1 Isolation Strategy

Phase 3 hits represent mixed communities. We need to identify which organism in each community is actually responsible for the activity. The strategy is iterative dilution-to-extinction combined with plating on selective media:

1. Streak each fluoride-positive culture onto multiple media types (rich, minimal, PFAS-supplemented minimal).
2. Pick 24–48 morphologically distinct colonies per culture.
3. Re-test each isolate individually in the Phase 3 fluoride assay.
4. For colonies that lose activity in pure culture, test pairwise and small-pool combinations — some degradation activity may require microbial consortia.

4.2 Genomic Characterization

Each confirmed-active isolate (and any active consortium) will be characterized by:

- **16S rRNA sequencing** for rapid taxonomic placement.
- **Whole-genome sequencing** (Illumina short-read + Oxford Nanopore long-read for hybrid assembly) of pure isolates.
- **Metagenomic sequencing** of consortia, with binning to recover individual genomes.
- **Targeted gene search** for known and putative defluorinases, haloacid dehalogenases, cytochrome P450s, and reductive dehalogenases.

4.3 Bioinformatics

Assemblies will be annotated with Prokka and compared against curated databases of known dehalogenation enzymes. Candidate novel defluorinases will be flagged for downstream heterologous expression in Phase 5 follow-on work or in subsequent grant cycles.

Go/No-Go criterion (end of Month 9): At least one pure isolate (or stable defined co-culture of ≤ 3 organisms) confirmed to release fluoride from PFAS, with a high-quality draft genome assembled.

5. Phase 5 — Proof-of-Concept Validation

Phase 5 Proof-of-Concept Validation *Months 9–12*

Characterize the isolated strain(s) under controlled laboratory conditions and confirm that PFAS is genuinely being degraded — not merely sorbed, transformed, or sequestered.

5.1 Mass Balance

The cleanest test of true degradation is a mass balance: input PFAS (mol C, mol F) should equal output PFAS + transformation products + released F^- + biomass-associated fluorine + mineralized product (CO_2 , if applicable). Deviations of more than 20% require investigation.

5.2 Validation Experiments

- **Dose response:** degradation rate vs. PFAS concentration (0.1–50 mg/L) to estimate kinetic parameters (K_m , V_{max}).
- **Substrate range:** test against a panel of 6–10 PFAS variants to map specificity (PFOA, PFOS, PFBA, PFBS, GenX, 6:2 FTS, etc.).
- **Cofactor and condition dependence:** O_2 vs. anoxic, temperature range, pH range, supplementary electron donors.
- **Transformation product identification:** high-resolution LC-MS/MS to identify intermediates and confirm whether degradation is partial or complete.
- **Sorption controls:** heat-killed cells exposed to PFAS measure abiotic sorption; true degradation is the difference between live and killed conditions.

5.3 Final Deliverables

- One or more fully characterized PFAS-degrading bacterial strains, deposited in the Stanford strain collection and (where appropriate) ATCC.
- Annotated genome assemblies for each isolate.
- A degradation kinetics dataset across a panel of PFAS substrates.
- A manuscript draft suitable for submission to a peer-reviewed journal.
- A go-forward plan for engineering improved variants and scale-up testing.

Final success criterion: At least one isolate demonstrates statistically significant, mass-balanced PFAS degradation with confirmed fluoride release across multiple PFAS substrates under defined laboratory conditions.

6. Timeline Summary

Phase	Activity		Months	Primary Output
1	Environmental Sampling	Sam-	1–2	4–6 sample sets, archived at 4°C and –80°C
2	Biological Mechanism		2–4	Enrichment cultures with documented C–F cleavage
3	High-Throughput Screening		4–6	Confirmed fluoride-positive culture wells
4	Isolation & Genomics		6–9	Pure isolates with sequenced genomes
5	Proof-of-Concept Validation	Vali-	9–12	Characterized strain(s), draft manuscript

7. Risk Register and Mitigations

Risk	Likelihood	Mitigation
No fluoride release in any sample	Low–Med	Sample 6 sites rather than 4; include co-metabolism arm; consider anoxic conditions in Phase 2 extension.
Activity lost on isolation	Medium	Test consortia in addition to pure cultures; maintain original mixed enrichment for re-derivation.
Sorption mistaken for degradation	Medium	Mass balance and heat-killed sorption controls in every phase; fluoride is a degradation-specific readout.
Hits are slow-growing or unculturable	Medium	Dilution-to-extinction and metagenome-binning give a path to genomic characterization even without isolation.
PFAS analytical bottleneck	Low–Med	Use the colorimetric F [–] assay as primary screen; reserve LC-MS/MS for confirmation only.

8. Personnel and Roles

- **Principal Investigator ():** scientific oversight, manuscript preparation.
- **Postdoctoral Scientist:** day-to-day experimental design, isolation, and genomics.
- **Research Technician:** sampling logistics, plate-based screening, sample logistics.

- **Bioinformatics Collaborator:** genome assembly, annotation, defluorinase candidate gene calling.
- **PFASolve (external):** site identification, baseline contamination data.

9. Budget Categories (high level)

- Personnel (postdoc, technician partial): largest line item.
- LC-MS/MS time (Phase 1 baseline + Phase 2/5 confirmation).
- Sequencing (Illumina short-read + Nanopore long-read for ~10 isolates plus metagenomes).
- Consumables (media, PFAS standards, fluoride assay reagents, plasticware).
- Sampling travel and shipping.

Closing note. The signature of a successful 12-month outcome is not a perfectly characterized strain ready for deployment — that is a multi-year engineering effort. It is a *validated, sequenced organism* (or defined consortium) that genuinely cleaves C–F bonds in PFAS, providing a defensible biological starting point for follow-on engineering, scale-up, and field-relevant testing.